Revo Air

Passenger Satisfaction Performance Report

Insights & Priorities Based on Passenger Feedback

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Executive Summary

1 Project Description

The airline operates multiple service and operational initiatives but lacks data-driven insight into which factors have the greatest impact on passenger satisfaction. This absence of prioritization constrains effective resource allocation and slows measurable customer experience improvements over the next six months

2 Objectives

- Identify the primary drivers of passenger satisfaction across operational, service, comfort, and digital experience dimensions
- Quantify satisfaction gaps between satisfied and neutral/dissatisfied passengers
- Support strategic decision-making by highlighting high-impact improvement areas

3 Tools

- **Python (Google Colab)** – data cleaning, aggregation, and metric calculations
- **Tableau Public** – interactive dashboards and visual analysis

4 Metrics

- Overall Satisfaction Rate
- Total Passenger Volume
- Average Driver Scores
- Satisfaction Gap

5 Analysis Result

The analysis shows that **Digital Experience** and **Physical Comfort** exhibit the **largest satisfaction gaps**, indicating these drivers have the **greatest influence** on whether passengers become satisfied or dissatisfied.
In contrast, **Operational Reliability** demonstrates a smaller gap, suggesting relatively stable performance with lower marginal impact on satisfaction improvement.

6 Recommendation

- Prioritize improvements in **Digital Experience** (online booking, boarding process, WiFi, in-flight entertainment) due to its **high satisfaction gap**
- **Maintain and incrementally** improve **Physical Comfort** and **Service Quality** to sustain satisfaction level
- **Focus Operational Reliability** efforts on specific pain points rather than broad initiatives
- **Use passenger segmentation** insights to design targeted service enhancements for **high-impact customer groups**

Methodology

This analysis follows a structured data analytics workflow—from problem framing to insight generation—using statistical analysis and visualization to support decision-making.

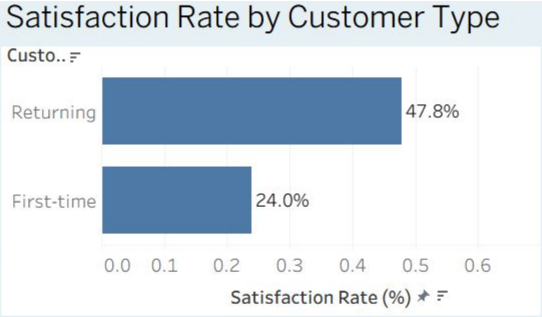
1. **Problem Understanding**
2. **Data Preparation & Data Cleaning**
3. **Exploratory Data Analysis**
4. **Data Analysis**
5. **Data Visualization**
6. **Insight & Recommendation**

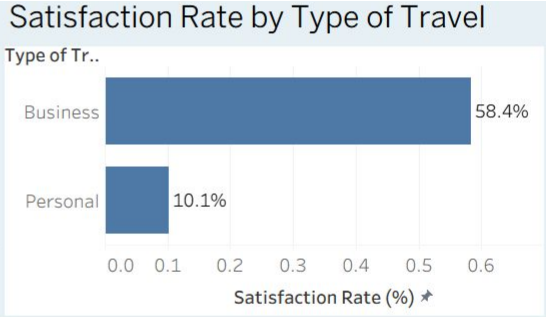
Overall Passenger Satisfaction Overview

Total Passenger	Overall Satisfaction	Overall At-Risk
129,487	43.4%	56.6%

- The airline serves **129,487 passengers**, indicating a large and diverse customer base.
- Only **43.4% of passengers are satisfied**, while **56.6% are neutral or dissatisfied**, meaning **more than half of passengers are at risk**.
- This imbalance highlights a **significant opportunity to improve customer experience** and reduce dissatisfaction at scale.

Satisfaction by Key Segments





Satisfaction by Travel Class

- **Business class passengers show the highest satisfaction (69.4%)**, significantly outperforming other classes.
- **Economy Plus satisfaction drops sharply to 24.6%**, and **Economy is the lowest at 18.8%**.
- This indicates a **strong service quality gap between premium and non-premium cabins**.

Economy and Economy Plus passengers represent the **largest dissatisfaction risk** and the **biggest opportunity for improvement**.

Satisfaction by Customer Type

- **Returning customers are almost twice as satisfied (47.8%)** compared to **first-time passengers (24.0%)**.
- First-time passengers are more likely to experience unmet expectations during their initial journey.

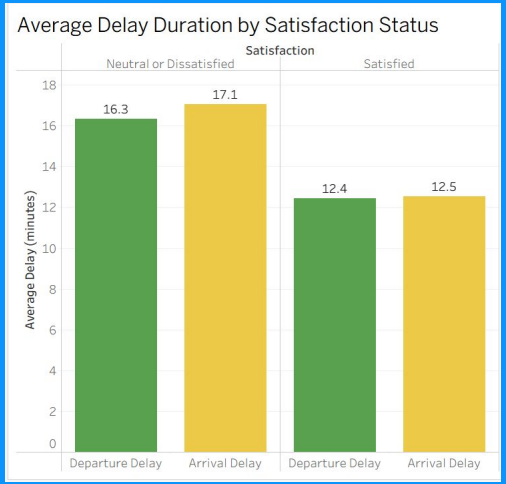
Early experience is critical — **poor first impressions may reduce customer retention**.

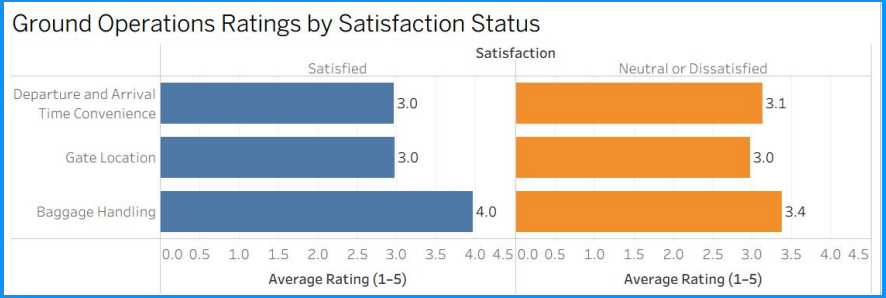
Satisfaction by Type of Travel

- **Business travel satisfaction is high (58.4%)**, while **personal travel satisfaction is very low (10.1%)**.
- Leisure travelers appear to be more price-sensitive and more affected by service inconsistencies.

Service delivery may be **optimized for business travelers**, but **fails to meet expectations for personal/leisure travelers**.

Operational Reliability Impact





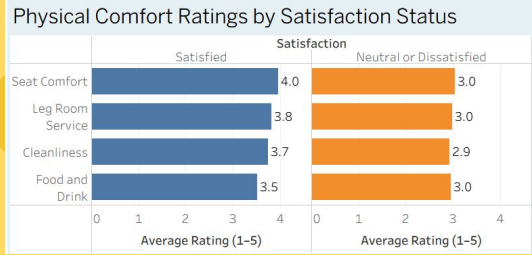
Key Analysis

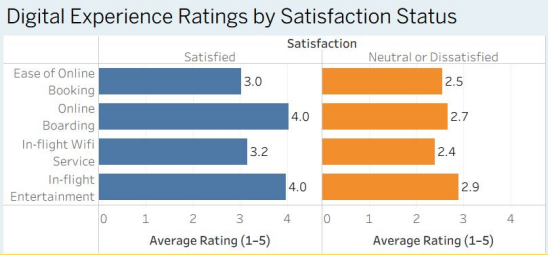
1. **Delay duration has a strong relationship with dissatisfaction**
   a. Passengers who are **Neutral/Dissatisfied experience longer delays:**
      - Departure delay: **~16.3 min vs ~12.4 min** (Satisfied)
      - Arrival delay: **~17.1 min vs ~12.5 min** (Satisfied)
   b. This shows that **even moderate increases in delay duration significantly affect satisfaction perception.**
2. **Arrival delays slightly worse than departure delays**
   a. Arrival delays are consistently **longer than departure delays** for dissatisfied passengers.
   b. This suggests **end-of-journey experience weighs more heavily on satisfaction.**
3. **Ground operations ratings are consistently lower for dissatisfied passengers**
   a. **Baggage handling** shows the largest gap:
      - Satisfied: **~4.0**
      - Neutral/Dissatisfied: **~3.4**
   b. **Time convenience & gate location** remain average (~3.0) but **do not compensate for delays.**

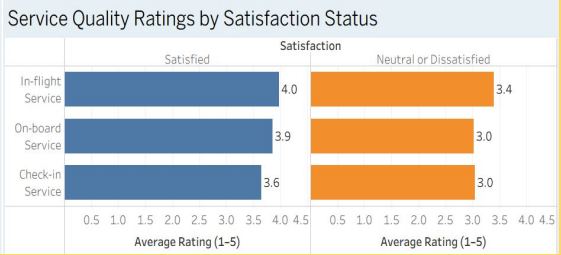
Key Insights

1. **Operational reliability is a primary driver of dissatisfaction**
2. **Arrival experience is more emotionally impactful**
3. **Baggage handling amplifies dissatisfaction when delays occur**

Service, Comfort & Digital Experience Drivers







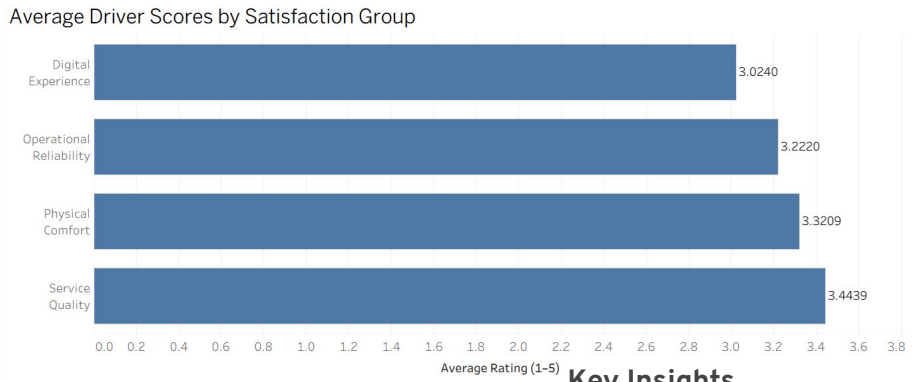
Key Analysis

- Across all three drivers, **satisfied passengers consistently rate experiences ~0.7–1.0 points higher**.
- **Digital Experience shows the largest gaps**, especially:
  - Online boarding
  - In-flight entertainment
  - WiFi
- **Physical comfort gaps** are notable in seat comfort and legroom.
- **Service quality gaps** exist across check-in, onboard, and in-flight service.

Insights

- Dissatisfaction is driven by **end-to-end experience gaps**, not a single touchpoint.
- **Digital experience is the most differentiating factor** between satisfied and dissatisfied passengers.
- Improvements in **service execution and comfort** can amplify satisfaction once operational basics are met.

Average Driver Performance



Key Analysis

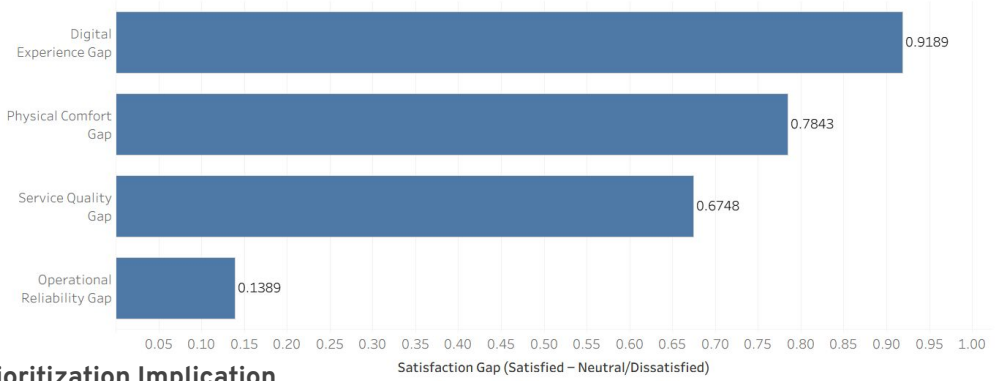
- There is a **clear performance gap** between *Satisfied* and *Neutral/Dissatisfied* passengers across all four driver groups.
- **Digital Experience** shows the **largest satisfaction gap**, indicating it is the most differentiating factor between satisfied and dissatisfied passengers.
- **Physical Comfort** and **Service Quality** also exhibit **substantial gaps**, suggesting inconsistent delivery of the core in-flight experience.
- **Operational Reliability** has the **smallest gap**, implying it is a baseline expectation: when it fails, dissatisfaction increases, but improving it alone is not enough to create high satisfaction.

Key Insights

- **Digital Experience is the strongest lever** for improving passenger satisfaction, as small improvements can significantly shift passengers from dissatisfied to satisfied.
- **Comfort and Service Quality are secondary but critical drivers**, reinforcing that in-flight experience consistency matters more than operational basics.
- **Operational Reliability acts as a hygiene factor**: it prevents dissatisfaction but does not strongly differentiate delight.
- Prioritizing investments based on **gap size** enables more effective resource allocation and faster impact on overall satisfaction metrics.

Driver Gap & Prioritization

Satisfaction Gap by Driver



Prioritization Implication

- **Digital Experience (Top Priority)**
  - Online booking
  - Boarding process
  - In-flight WiFi & entertainment
- **Physical Comfort (High Priority)**
  - Seat comfort & legroom
  - Cabin cleanliness
  - Food & beverage quality
- **Service Quality (Secondary Priority)**
  - In-flight and on-board service consistency
- **Operational Reliability (Maintenance Priority)**
  - Continue managing delays, but not the primary lever for satisfaction uplift

Key Analysis

- **Digital Experience shows the largest satisfaction gap (~0.92)**
  → The biggest difference between satisfied and at-risk passengers occurs in digital touchpoints.
- **Physical Comfort is the second-largest gap (~0.78)**
  → Comfort-related factors (seat, legroom, cleanliness, food) strongly separate satisfied vs dissatisfied passengers.
- **Service Quality shows a moderate gap (~0.67)**
  → Service execution matters, but its impact is weaker than Digital and Comfort.
- **Operational Reliability has the smallest gap (~0.14)**
  → Delays and ground operations affect satisfaction, but they are **less differentiating** between satisfied and dissatisfied groups.

Recommendations

Digital Experience (Top Priority)

O – Outcome
Increase overall passenger satisfaction and reduce at-risk passengers.

B – Business Insight
Digital experience is the **largest driver gap**, indicating the strongest opportunity to improve satisfaction through targeted investment.

I – Insight
Satisfied passengers rate online boarding, booking convenience, WiFi, and in-flight entertainment **~0.9 points higher** than neutral/dissatisfied passengers — the widest gap across all drivers.

P – Problem
Current digital touchpoints create friction that disproportionately impacts dissatisfied passengers.

R – Recommendation

- Simplify online booking and boarding flow
- Improve in-flight WiFi reliability
- Enhance usability and content quality of in-flight entertainment

Physical Comfort (High Priority)

O – Outcome
Improve comfort-related satisfaction, especially for economy-class passengers.

B – Business Insight
Physical comfort is the **second-largest satisfaction gap**, making it a key contributor to dissatisfaction.

I – Insight
Seat comfort, legroom, cleanliness, and food & beverage ratings are consistently **lower among dissatisfied passengers**, with a gap of ~0.78 points.

P – Problem
Inconsistent comfort standards reduce perceived value, particularly on longer flights and non-business classes.

R – Recommendation

- Improve seat comfort and legroom optimization
- Strengthen cabin cleanliness standards
- Enhance food and beverage consistency

Recommendations

Service Quality (Medium Priority)

O – Outcome
Increase consistency of service experience across all customer touchpoints.

B – Business Insight
Service quality acts as a **supporting driver** that reinforces satisfaction when combined with comfort and digital improvements.

I – Insight
Check-in, on-board service, and in-flight service show a **moderate satisfaction gap (~0.67)** between satisfied and dissatisfied passengers.

P – Problem
Variability in service delivery leads to uneven passenger experiences.

R – Recommendation

- Standardize service procedures
- Enhance frontline staff training
- Reinforce service quality monitoring

Operational Reliability (Maintain & Monitor)

O – Outcome
Maintain baseline satisfaction by preventing operational issues from escalating dissatisfaction.

B – Business Insight
Operational reliability has the **smallest satisfaction gap**, indicating it is a hygiene factor rather than a differentiator.

I – Insight
Delay duration differs between satisfied and dissatisfied passengers, but the gap (~0.14) is significantly smaller than other drivers.

P – Problem
While not a primary driver, poor reliability can quickly erode trust if performance declines.

R – Recommendation

- Maintain current delay management performance
- Improve communication transparency during delays

Thanks!

Dataset

Link Google Colab

Link Tableu

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